# A khronon-split bimetric host for the dark-energy–anchored acceleration scale: construction, constraint structure, and the ghost test

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We construct a bimetric completion of MOND phenomenology carrying the acceleration-scale identity $a_0^2 = \kappa^2 G(-K(\mathcal{Q}))$ — the MOND scale as the dark sector's pressure — and push it through every consistency gate our previously published filter [10.5281/zenodo.22004372, 10.5281/zenodo.22015358] can pose. The construction: two Einstein–Hilbert sectors; a shift-symmetric khronon $\phi$ whose $\beta = 1$ DBI kernel supplies dark energy ($w = -1$ exactly), a cold dust component, and a derived $a_0(z)$; and an interaction built from the khronon-projected connection difference $C_M{}^i{}_{jk} = h^i{}_\mu h_j{}^\nu h_k{}^\rho\, C^\mu{}_{\nu\rho}$, entering as $V = \sqrt{h}\,[NF(X) + \hat{N}B(X)]$ with $X$ the projected quadratic scalar normalised by $a_0^2(\mathcal{Q})$. Results, each with a committed self-checking script: **(1)** after the Hassan–Rosen-type shift redefinition $u^i = (N^i - \hat{N}^i)/\hat{N}$, $C_M$ is exactly lapse-free, and the interaction's lapse Hessian vanishes identically — all four entries. **(2)** The quasi-static reduction of a bimetric theory is a two-potential system whose force law is a Möbius map in $F'$, not AQUAL; inverting it for the $a_0$-line $g_{\rm obs}^2 = g_{\rm bar}^2 + a_0 g_{\rm bar}$ yields the required interaction slope in closed form, $\hat{\alpha}F' = (1 - R)/(R(1 + r) - 1)$ with $R = (\sqrt{1 + 4x^2} + 1)/2x$, and it is single-valued and monotone: the legality condition that excludes the same phenomenology in aether–scalar–tensor theory is satisfied here. The construction's free content is one function fixed by the $a_0$-line plus one number, $r = M_f^2/M_g^2$. **(3)** The sign that makes gravity stronger rather than weaker is derived, not chosen: calibrating the Einstein–Hilbert quadratic coefficient against the Newtonian limit ($M^2 = 1/8\pi G$ recovered exactly) forces the *mixed* contraction $C^i{}_{jk}C^j{}_{ik} = -|\nabla\psi|^2$. **(4)** Both lapses appear linearly in the Hamiltonian; $u^i$ is determined algebraically, absorbing one momentum constraint; the degree-of-freedom count is $7 = 2 + 5$, the ghost-free bimetric number, conditional on one bracket. **(5)** That bracket is tested: $\{F(X), B(X)\} = 0$ identically (two functions of the same phase-space scalar), collapsing $\{C, \hat{C}\}$ to two cross terms; and on an explicit lattice representative, with every constraint imposed at every site and both controls passing exactly, the on-shell bracket is *nonzero* at a generic real point. First-classness is universal, so the ghost-propagating alternative is refuted on the representative: the pair is second class and the Boulware–Deser mode is removed there. **Not claimed:** closure. The exact kinetic reduction, the continuum statement, and full Dirac consistency remain, and are listed. Every prior error made in this programme — six, roughly half manufactured deficits and half manufactured wins — is recorded, dated, with directions stated, in the repository's retraction log.

## I. THE CONSTRUCTION

The action:

$$S = \frac{M_g^2}{2}\int\sqrt{-g}\,R[g] + \frac{M_f^2}{2}\int\sqrt{-\hat{g}}\,\hat{R}[\hat{g}] + S_\phi[\phi, g] - m^2 M_{\rm eff}^2\int N\sqrt{h}\Big[F(X) + \tfrac{\hat{N}}{N}B(X)\Big] + S_{\rm m}[g, \Psi], \quad (1)$$

with $1/M_{\rm eff}^2 = 1/M_g^2 + 1/M_f^2$ and matter coupled to $g_{\mu\nu}$ alone. The khronon sector $S_\phi$ carries $\mathcal{Q} = \sqrt{-g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi}$ and the $\beta = 1$ DBI kernel $K(\mathcal{Q}) = -\rho_\Lambda c^2\sqrt{1 - (\mathcal{Q} - \mathcal{Q}_0)^2/\Lambda_D^2}$, giving $w = -1$ exactly at the minimum, dust excitations carrying the dark abundance as a conserved shift charge, and — via the promotion

$$a_0^2(\mathcal{Q}) = \kappa^2 G\,(-K(\mathcal{Q})), \quad (2)$$

the identity that is this programme's contribution — the derived evolution $a_0(a)/a_0(0) = (1+\sigma^2)^{-1/4}$, $\sigma \propto \nu_0/a^3$. At the minimum $a_0 = \kappa c\sqrt{G\rho_\Lambda} = 9.3619 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ (alternative footing $1.1279\times10^{-10}$; both carried throughout). $\kappa = \frac{1}{2}$ is *fitted*, $0.529 \pm 0.034$, never derived. The $a_0$–$\Lambda$ *coincidence* is prior art [5–8]; the identity (2) and its consequences are not.

The interaction's scalar is built from the relative connection $C^\mu{}_{\nu\rho} = \Gamma^\mu{}_{\nu\rho}[g] - \hat{\Gamma}^\mu{}_{\nu\rho}[\hat{g}]$ (a tensor, as in bimetric MOND [1]), projected fully spatially with the khronon's foliation $n_\mu = \partial_\mu\phi/\sqrt{-(\partial\phi)^2}$, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$: $X$ is the *mixed* quadratic contraction of $C_M$, normalised by $a_0^2(\mathcal{Q})$. The foliation is a dynamical field, not a gauge choice; full diffeomorphism invariance is kept.

## II. FIVE RESULTS

*1. Lapse-freeness.* In unitary gauge, the physical lapse $N$ cancels from $C_M$ identically at the projection stage, and the residual $\hat{K}_{jk}(\hat{N}^i - N^i)/\hat{N}$ is absorbed by the invertible redefinition $u^i = (N^i - \hat{N}^i)/\hat{N}$:

$$C_M{}^i{}_{jk} = \left({}^3\Gamma^i{}_{jk} - {}^3\hat{\Gamma}^i{}_{jk}\right) - \hat{K}_{jk}u^i, \quad (3)$$

free of both lapses. The interaction's lapse Hessian then vanishes *identically* — all four entries, not a diagonal zero (twice in this programme a diagonal zero was mistaken for degeneracy; $\det H = -(\text{mixed})^2$ punishes that mistake).

*2. The force law, and the function it fixes.* Matter couples to $g$ alone, so the hatted potential equation is sourceless and the quasi-static reduction is a Möbius map, $g_{\rm obs}/g_{\rm bar} = [1+\hat\alpha F']/[1+(1+r)\hat\alpha F']$, with $\alpha/\hat\alpha = M_f^2/M_g^2 \equiv r$ exactly (each sector divides the same interaction flux by its own Planck mass). Inverting for the $a_0$-line gives, in closed form,

$$\hat\alpha F'(x) = \frac{1-R}{R(1+r)-1}, \qquad R(x) = \frac{\sqrt{1+4x^2}+1}{2x}, \quad (4)$$

$x = g_{\rm obs}/a_0$. It is negative-definite, monotone, and single-valued in $X$; it vanishes in the Newtonian limit (the interaction switches itself off — solar-system screening from the required force law, not from a kernel choice) and tends to $-1/(1+r)$ in deep MOND. Only the product $\hat\alpha F'$ is observable, so the interaction strength is degenerate with $F$'s normalisation: the construction's free content is Eq. (4) plus the single number $r$.

*3. The sign, calibrated.* Computing $\sqrt{-g}R$ at quadratic order and matching the Newtonian limit recovers $M^2 = 1/8\pi G$ exactly; carrying that calibration through the variation gives $\mathrm{sign}(\alpha) = \mathrm{sign}(c)$ for contraction coefficient $c$, and MOND (enhancement, not weakening) requires $\alpha < 0$: the mixed contraction $C^i{}_{jk}C^j{}_{ik} = -|\nabla\psi|^2$ is forced. No negative coupling is inserted by hand. (The sign was asserted wrongly once and reversed once before being calibrated; both are in the retraction log. The lesson — a coefficient asserted is a coefficient wrong — is now a standing rule.)

*4. Constraint structure and the count.* Substituting the redefinition, both lapses appear linearly: $H = \int N\,C + \hat N\,\hat C + \hat N^i(\mathcal{H}_i + \hat{\mathcal{H}}_i)$ with $C = \mathcal{H}_{\rm EH} + \sqrt{h}F(X)$ and $\hat C = \hat{\mathcal{H}}_{\rm EH} + \sqrt{\hat h}B(X) + u^i\mathcal{H}_i$. The relative shift $u^i$ is determined algebraically (its equation is linear), absorbing one momentum constraint — the Hassan–Rosen shape [2]. Counting: $24-6-2-2 = 14$ phase-space dimensions, i.e. $7 = 2+5$ degrees of freedom, the ghost-free bimetric number — *provided* the orthogonal combination of $(C,\hat C)$ is second class. If instead both were first class the count is 8, and the eighth is the Boulware–Deser ghost.

*5. The ghost test.* Two of the four terms of $\{C,\hat C\}$ vanish identically: the two Einstein–Hilbert constraints commute (disjoint pairs), and $\{F(X), B(X)\} = 0$ for *any* $X$ — two functions of the same phase-space scalar Poisson-commute, a one-line theorem that survived the withdrawal of its own earlier, wrong proof. The two survivors are cross terms, each one sector's Einstein–Hilbert constraint against the other sector's interaction piece. On an explicit representative — anisotropic minisuperspace on a three-site lattice, exact rational arithmetic — with *every* constraint imposed at *every* site and the surface reached through the free function's own values (which enter linearly; no branch ambiguity; all nine remaining momenta generic reals), the on-shell bracket evaluates to $-7.35 \neq 0$. Both controls pass exactly: interaction off $\Rightarrow$ bracket $\equiv 0$; $F, B$ constant $\Rightarrow$ bracket $\equiv 0$. Since first-classness must hold at every phase-space point, one generic point of failure refutes it: *on the representative the pair is second class, the Boulware–Deser mode is removed, and the count of five results stands at 7.*

## III. WHAT REMAINS, AND WHAT IS CLAIMED

*Discharged since the ghost test:* the exact ADM kinetic reduction. Deriving $H_{\rm kin} = -p_a p_b/4b + a p_a^2/8b^2$ by exact Legendre transform (no $p_b^2$ term — the declared form had one) and the exact $\sqrt{h}R^{(3)}$ potential, the on-shell bracket remains nonzero: the verdict does not rest on the declared coefficients. *Remaining, one honest and one adverse-leaning:* the continuum statement (a representative refutes universal first-classness but cannot prove general second-classness); and Dirac termination — on the representative the $6\times 6$ on-shell multiplier matrix $\Delta_{IJ} = \{C_I, C_J\}$ has full rank, which naively fixes *every* lapse combination including the diagonal time-reparametrisation the diffeomorphism invariance of (1) guarantees. That reads as the frozen-lapse situation and is reported as found; the suspect is the three-site open-boundary lattice (the diagonal combination's first-classness lives in derivative-of-delta closure on the momentum constraint, which such a lattice breaks), but that explanation is *not verified* and the continuum Dirac analysis is the remaining theorem-grade item. *Remaining on the physics fronts, unchanged:* a Boltzmann computation for the coupled system; lensing $\Phi + \Psi$ in the combined weak-field limit; clusters; and whether the dark sector's dust remains bound in galaxies.

*Observational shape carried over:* the derived $a_0(z)$ predicts an essentially static baryonic Tully–Fisher normalisation across $0 < z \lesssim 20$ (velocity shifts of $10^{-9}$–$10^{-6}$ at $z = 1$–$5$) followed by a cliff (1% at $z \approx 26$, 10% at $z \approx 48$) — a shape no constant-$a_0$ or smoothly-evolving-$a_0$ model has, and a two-sided one: a robust per-cent-level detection of $a_0$ evolution below $z \sim 5$ would require $\nu_0$ three to four orders of magnitude above its independent ceilings and falsifies the law outright.

*Claimed:* the construction; the five results; and that this is the first candidate in this programme — after a published three-condition filter and the recorded deaths of every predecessor — to pass every gate it has faced, including the ghost test that killed the rest. *Not claimed:* a closed theory; a derived $\kappa$; or the absence of dark matter — the dark abundance is present in full as a field's conserved charge, and both the $w = -1$ result and the CMB depend on it.

## ACKNOWLEDGMENTS

Every quantitative claim is reproduced by committed self-checking scripts at https://github.com/carlzimmerman/zimmerman-formula (`qwen_claude_field_theory/closure_2026/`), each printing numbered checks and exiting non-zero on failure. Six errors made during this programme are recorded, dated, with directions stated, in `RETRACTIONS.md`; roughly half manufactured deficits, half manufactured wins, and the standing rules extracted from them (check the full Hessian; calibrate every coefficient against a control; a result proved in one regime may not be carried into another; matching limits is not matching a function) are stated in the repository. The underlying theories are Milgrom's (BIMOND, QUMOND) and Hassan–Rosen's constraint mechanism; the previous host analysis is Skordis–Złośnik's AeST. The contribution is the promotion (2), the khronon-projected split, the closed-form (4), and the gate results.



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